Degraded performance of EarthScope prod hub

Status: Reviewed

Overview

During a cloud computing and parallelization for seismologists workshop, attendees experienced degraded server performance following the onset of high volume disk writing

What Happened

~30 attendees of the workshop ran parallelised analysis routines. These routines performed high-volume IO to the home directory storage. Subsequently, the EBS disk backing the central NFS server reached its provisioned throughput limit, leading to iowait on the core node CPU. This led to user servers experiencing severe delays in FS operations on home directories, which manifested as degraded user experience in the JupyterLab interface and kernel execution.

Resolution

Once throughput demand dropped below the provisioned value, the iowait on the core node dropped to a nominal value.

Where we got lucky

No comments added

What Went Well?

We had an alert to detect large iowait incidences.

What Didn't Go So Well?

The iowait saturation alert took 30m to trigger a PagerDuty incident from the onset of the incident.
The incident occurred outside of regular business hours (in the late evening) for the on-staff engineers, meaning that they didn't respond in realtime.
The iowait alerts were not considered alarming when retrospectively reviewing the alerts the following day.

Action Items

- https://github.com/2i2c-org/infrastructure/issues/8128
- https://github.com/2i2c-org/infrastructure/issues/8129
- https://github.com/2i2c-org/infrastructure/issues/8130
- https://github.com/2i2c-org/infrastructure/issues/8131
- https://github.com/2i2c-org/infrastructure/issues/8132
- https://github.com/2i2c-org/infrastructure/issues/6249
- https://github.com/2i2c-org/infrastructure/issues/8133
- https://github.com/2i2c-org/infrastructure/issues/8221

Timeline

OWNER OF REVIEW PROCESS

Angus

IMPACT TIME

Apr 14 at 23:00 to Apr 17 at 01:59

DURATION

2d 2h 59m

*All times listed in this report are in Bucharest.

Apr 14, 2026

11:05 PM High volume disk writing starts on the home-directory disk

11:38 PM First alert of iowait saturation

Alert [FIRING:1] Disk IO approaching saturation earthscope (nvme2n1 ip-192-168-5-182.us-east-2.compute.internal action needed this week) was automatically added to this incident.

INCIDENT #2103

[FIRING:1] Disk IO approaching saturation earthscope (nvme2n1 ip-192-168-5-182.us-east-2.compute.internal action needed this week)

Apr 15, 2026

12:06 AM Second alert of iowait saturation

Alert [FIRING:1] Disk IO approaching saturation earthscope (nvme2n1 ip-192-168-5-182.us-east-2.compute.internal action needed this week) was automatically added to this incident.

INCIDENT #2104

[FIRING:1] Disk IO approaching saturation earthscope (nvme2n1 ip-192-168-5-182.us-east-2.compute.internal action needed this week)

1:31 AM Alert resolves after iowait stabilised

Resolved through the integration API.

INCIDENT #2104

[FIRING:1] Disk IO approaching saturation earthscope (nvme2n1 ip-192-168-5-182.us-east-2.compute.internal action needed this week)